

Vector Space

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Binary Operations

What is a binary operation?

Binary Operations

Definition

Any function from $A \times A \rightarrow A$ is a binary operation.

- **Closure Law:**
  - A set is said to be closure under an operation (like addition, subtraction, multiplication, etc.) if that operation is performed on elements of that set and result also lies in set.

$$if\ a \in A, b \in A \rightarrow a * b \in A$$

03

Vector Space

Vector Space

- Building blocks of linear algebra.
- A non-empty set V with field F (most of time R or C) forms a vector space with two operations:
  1. + : Binary operation on V which is V x V → V
  2. . : F x V → V

Note

In our course, by **default**, **field** is **R** (real numbers).

Vector Space

Definition

A vector space over a field F is the set V equipped with two operations: $(V, F, +, .)$

i. Vector addition: denoted by "+" adds two elements $x, y \in V$ to produce another element $x + y \in V$

ii. Scalar multiplication: denoted by "." multiplies a vector $x \in V$ with a scalar $\alpha \in F$ to produce another vector $\alpha . x \in V$. We usually omit the "." and simply write this vector as αx.

Vector Space Properties

- **Addition of vector space** $(x + y)$

  - **Commutative** $x + y = y + x \; \forall x, y \in V$
  - **Associative** $(x + y) + z = x + (y + z) \; \forall x, y, z \in V$
  - **Additive identity** $\exists \mathbf{0} \in V$ such that $x + \mathbf{0} = x, \forall x \in V$
  - **Additive inverse** $\exists (-x) \in V$ such that $x + (-x) = 0, \forall x \in V$

Vector Space Properties

- **Action of the scalars field on the vector space (αx)**
  - **Associative** $\alpha(\beta x) = (\alpha\beta)x$ $\forall \alpha, \beta \in F; \forall x \in V$
  - **Distributive over ……**

    scalar addition: $(\alpha + \beta)x = \alpha x + \beta x$ $\forall \alpha, \beta \in F; \forall x \in V$

    vector addition: $\alpha(x + y) = \alpha x + \alpha y$ $\forall \alpha \in F; \forall x, y \in V$
  - **Scalar identity** $1x = x$ $\forall x \in V$

Vector Space of functions

- Function addition and scalar multiplication

$$(f+g)(x) = f(x) + g(x) \quad and\ (af)(x) = af(x)$$

Non-empty set X and any field F $\rightarrow F^X = \{f : X \rightarrow F\}$

Example

- Set of all polynomials with real coefficients
- Set of all real-valued continuous function on [0,1]
- Set of all real-valued function that are differentiable on [0,1]

Vector Space of polynomials

$P_n\ (\mathbb{R})$: Polynomials with max degree (n)

- Vector addition
- Scalar multiplication
- And other 8 properties!

Vector Space

Example

Which are vector spaces with + , *?

- ❑ Set $\mathbb{R}^n$ over $\mathbb{R}$
- ❑ Set $\mathbb{C}$ over $\mathbb{R}$
- ❑ Set $\mathbb{R}$ over $\mathbb{C}$
- ❑ Set $\mathbb{Z}$ over $\mathbb{R}$
- ❑ Set of all polynomials with coefficient from $\mathbb{R}$ over $\mathbb{R}$
- ❑ Set of all polynomials of degree at most n with coefficient from $\mathbb{R}$ over $\mathbb{R}$
- ❑ Matrix: $M_{m,n}(\mathbb{R})$ over $\mathbb{R}$
- ❑ Function: $f(x): x \longrightarrow \mathbb{R}$ over $\mathbb{R}$

Vector Space

Example

Let V be the set of all real numbers with the operations $u \oplus v = u - v$, $\oplus$ is an ordinary subtraction) and $\alpha \odot u = \alpha u$ ($\odot$ is an ordinary multiplication).
Is V a vector space? If it's not, which properties fail to hold?

Resources

- Kenneth Hoffman and Ray A. Kunze. Linear Algebra. PHI Learning, 2004.
- David C. Lay, Steven R. Lay, and Judi J. McDonald. Linear Algebra and Its Applications. Pearson, 2016.
- Gilbert Strang. Introduction to Linear Algebra. Wellesley-Cambridge Press, 2016.